Assignment 1 - Git & GitHub Fundamentals

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1. Repository Setup

I initialized a Git repository on the main branch. The .gitignore file excludes local.env and normal .log files while !important.log keeps important.log visible to Git.

2. README & Commit History

My README includes my name, GitHub username, description, technologies, example commands, a documentation link, and the provided starter image. I used meaningful commit messages for the setup, README, learning goals, About feature, and selective-staging exercise.

3. Selective Staging

I modified README.md and project-notes.md at the same time. I staged and committed only README.md, leaving project-notes.md modified. I checked the changes using git status, git diff, and git diff --cached.

4. Branch & Merge

I created feature/about, added an About section to README.md, committed it on the feature branch, returned to main, and merged the feature branch.

5. Remote Changes

A commit records changes locally, while a push uploads commits to the remote repository. Fetch downloads remote updates without integrating them into the current branch; pull downloads and integrates remote changes. I verified that local main was one commit behind origin/main after fetching, then pulled the remote README update.

6. Fair Use of AI

I used ChatGPT to help organize the Git workflow, explain commands, and structure this report. I reviewed the commands and verified the results using Git status, diffs, commit history, and the fetch/pull exercise.

7. Sources & Reflection

Sources: Course assignment instructions and Git documentation.

The hardest concept for me was selective staging because staged and unstaged changes can exist simultaneously. If Git behaves unexpectedly, I would first check git status, the current branch, git remote -v, and git log --oneline --graph --all.